UNISA university of south africa

Student number: 6215-857-0
Date: 2018-09-28

MAZIBUKO X MR
1 LAVENDER GROVE GLOVER ROAD
ROODEPOORT
1724

Dear Student

As requested, a statement is attached.

Yours faithfully

Dr F Goolam

Registrar

UNIVERSITEIT VAN SUID-AFRIKA
PRETORIA
UNIVERSITY OF SOUTH AFRICA

University of South Africa
Preller Street, Muckleneuk Ridge, City of Tshwane
PO Box 392 UNISA 0003 South Africa
Telephone: +27 12 429 3111 Facsimile: +27 12 429 4150
www.unisa.ac.za

UNISA university of south africa

Student number: 6215-857-0
Date: 2018-09-28

This is to certify that

XOLANI MAZIBUKO

Identity Number :9210045040082
Date of Birth :1992-10-04

passed the university examinations in the undermentioned study units for which credit has been granted in partial completion of the

Bachelor of Science

NQF exit level: 7
Minimum credits required: 360

YEAR	MONTH	CODE	NAME OF STUDY UNIT	%	NQF LEVEL	CREDITS
2017	MAY	COS1501	* Theoretical Computer Science I	99	5	12
2017	JUN	COS1511	* Introduction to Programming I	75	5	12
2017	OCT	COS1512	* Introduction to Programming II	77	5	12
2017	MAY	COS1521	* Computer Systems: Fundamental Concepts	80	5	12
2017	OCT	COS2601	* Theoretical Computer Science II	81	6	12
2017	NOV	COS2661	* Formal Logic II	81	6	12
2017	JUN	EUP1501	Ethical Information and Communication Technologies for Development Solutions	58	5	12
2018	JUN	MAT2612	* Introduction to Discrete Mathematics	81	6	12
2018	MAY	PLS2607	Philosophy of Science	54	6	12
2018	JUN	COS2611	Programming: Data Structures	65	6	12
2018	JUN	COS3761	* Formal Logic III	78	7	12

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* Passed with distinction

Exemption from the following has been granted by virtue of corresponding study units passed:

2018	JUN	MAT1503	Linear Algebra		5	12
2018	JUN	MAT1512	Calculus A		5	12
2018	JUN	MAT1613	Calculus B		6	12
2018	JUN	MAT2611	Linear Algebra		6	12

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Total credits accumulated: 180

Major subject(s): COMPUTER SCIENCE
MATHEMATICS

This qualification is not completed.

Yours faithfully

Registrar

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Purpose statement of modules passed

This is to certify that the purpose statement of the modules offered comprises the following:

COS1501 - Theoretical Computer Science I

To introduce students to some concepts from Discrete Mathematics as a theoretical foundation for Computer Science. This background is relevant to relational databases, the development of provably correct programs, and the analysis of algorithms.

COS1511 - Introduction to Programming I

To provide students with an introduction to programming and to cover the fundamentals of control structures, problem-solving techniques, and the incremental testing of programs.

COS1512 - Introduction to Programming II

To introduce students to the detailed design and implementation of algorithms as programs, and includes the fundamentals of simple data structures with object-orientation.

COS1521 - Computer Systems: Fundamental Concepts

To introduce students to the computer as a system. This covers hardware concepts such as internal representation of numbers and characters and basic computer architecture, and software concepts such as systems software and applications software. It also includes a brief introduction to databases, and to systems analysis and design.

COS2601 - Theoretical Computer Science II

This module together with COS3701 will acquaint students with the capabilities and limitations of computers from a theoretical viewpoint. Module COS2601 covers formal languages, recursive definitions, regular expressions, finite automata, Moore and Mealy machines, transition graphs, the pumping lemma and decision problems.

COS2611 - Programming: Data Structures

To show learners how abstract data types and data structures can be implemented and used in an object-oriented programming language. The module covers recursion, linked lists, dynamic memory allocation, binary trees, and graphs.

COS2661 - Formal Logic II

To introduce students to the syntax of propositional language and the truth functionality of first order logic, enabling them to deal with proofs for validity via deduction and resolution in an interpreted first-order language.

COS3761 - Formal Logic III

To enable students to construct a number of different formal languages (such as opaque or transparent propositional languages, firstorder languages, sorted languages, modal languages and non-monotonic logics) and to solve problems of validity and truth for these languages.

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EUP1501 - Ethical Information and Communication Technologies for Development Solutions

Students who successfully complete this module will be able to present critical arguments around ICTs for development, while appreciating ethical dimensions within an information society in relation to copyright and intellectual property rights, privacy, conduct and expectations. In addition, successful students will be able to demonstrate their ability to maintain efficient, organised and secure electronic working environments by managing digital files, systems and application software, as well as capabilities related to engaging with textual and numerical data to present information in various formats.

MAT1503 - Linear Algebra

To enable students to understand and apply the following basic concepts in linear algebra: non-homogeneous and homogeneous systems of linear equations, Gaussian and Jordan-Gauss elimination, matrices and matrix operations, elementary determinants by cofactor expansion, inverse of matrix using the adjoint, Cramer's rule, evaluating determinants using row/column reduction, properties of the determinant function, vectors in 2- , 3- and n- space, dot product, projections, cross product, areas of parallelograms and volumes of parallelepipeds determined by vectors, lines and planes in 3-space and complex numbers.

MAT1512 - Calculus A

To equip students with those basic skills in differential and integral calculus which are essential for the physical, life and economic sciences. Some simple applications are covered. More advanced techniques and further applications are dealt with in module MAT1613.

MAT1613 - Calculus B

To enable students to obtain basic skills in differentiation and integration, and build on the knowledge provided by module MAT1512. More advanced techniques and further basic applications are covered. Together, the modules MAT1512 and MAT1613 constitute a first course in Calculus which is essential for students taking Mathematics as a major subject

MAT2611 - Linear Algebra

To understand and apply the following linear algebra concepts:vector spaces, rank of a matrix, eigenvalues and eigenvectors, diagonalisation of matrices, orthogonality in Rn, Gram-Schmidt algorithm, orthogonal diagonalisation of symmetric matrices, linear transformations, change of basis and their matrix representations.

MAT2612 - Introduction to Discrete Mathematics

To acquaint students with the theory and applications of the following aspects of discrete mathematics: counting principles, relations and digraphs, (including equivalence relations), functions, the pigeonhole principle, order relations and structures (e.g. partially ordered sets, lattices, Boolean algebras), the principle of induction.

PLS2607 - Philosophy of Science

To acquaint students with the nature of scientific reasoning, the status of scientific theories in terms of their relation(s) to reality, and connections between the theories and practice of science.

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